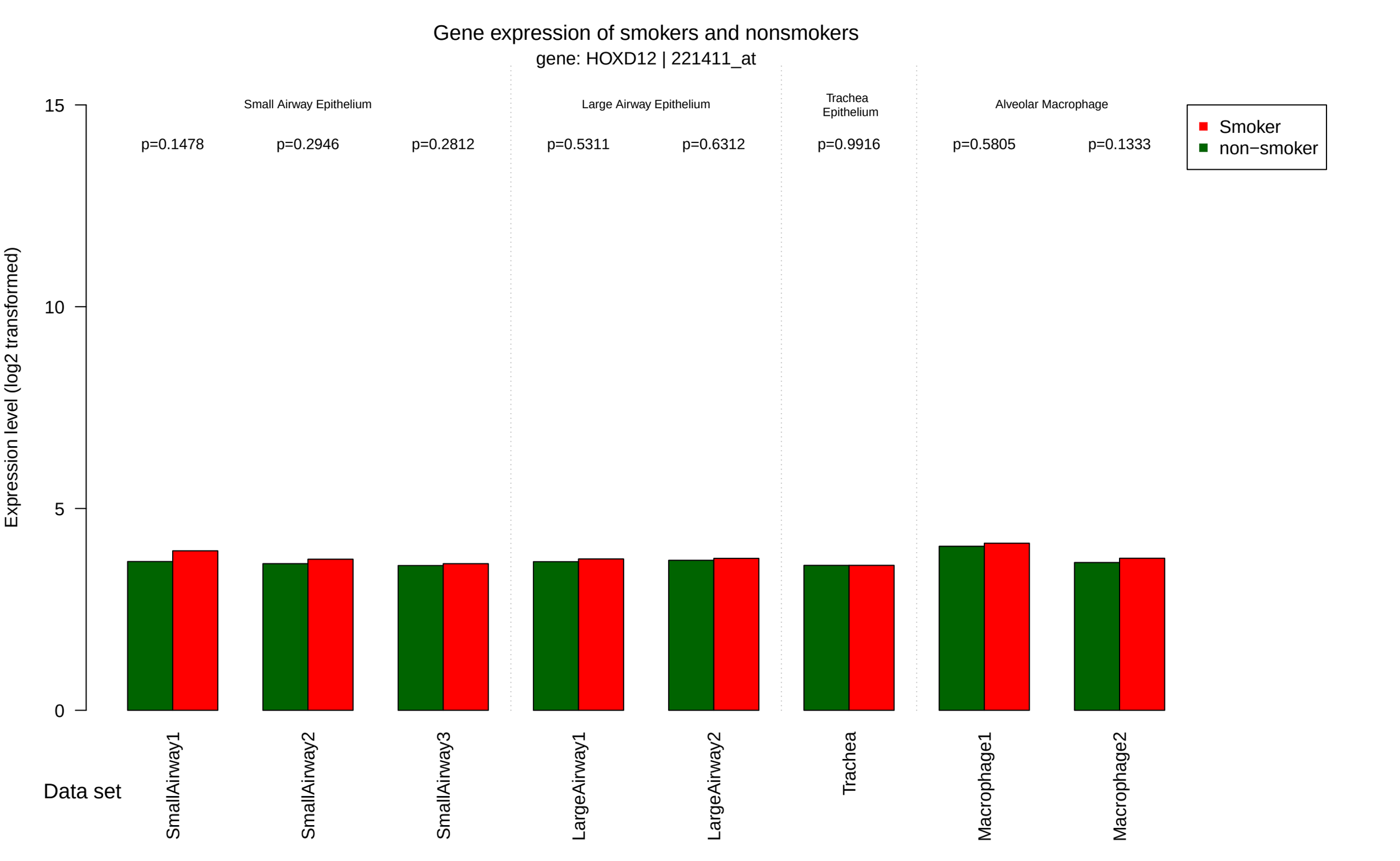
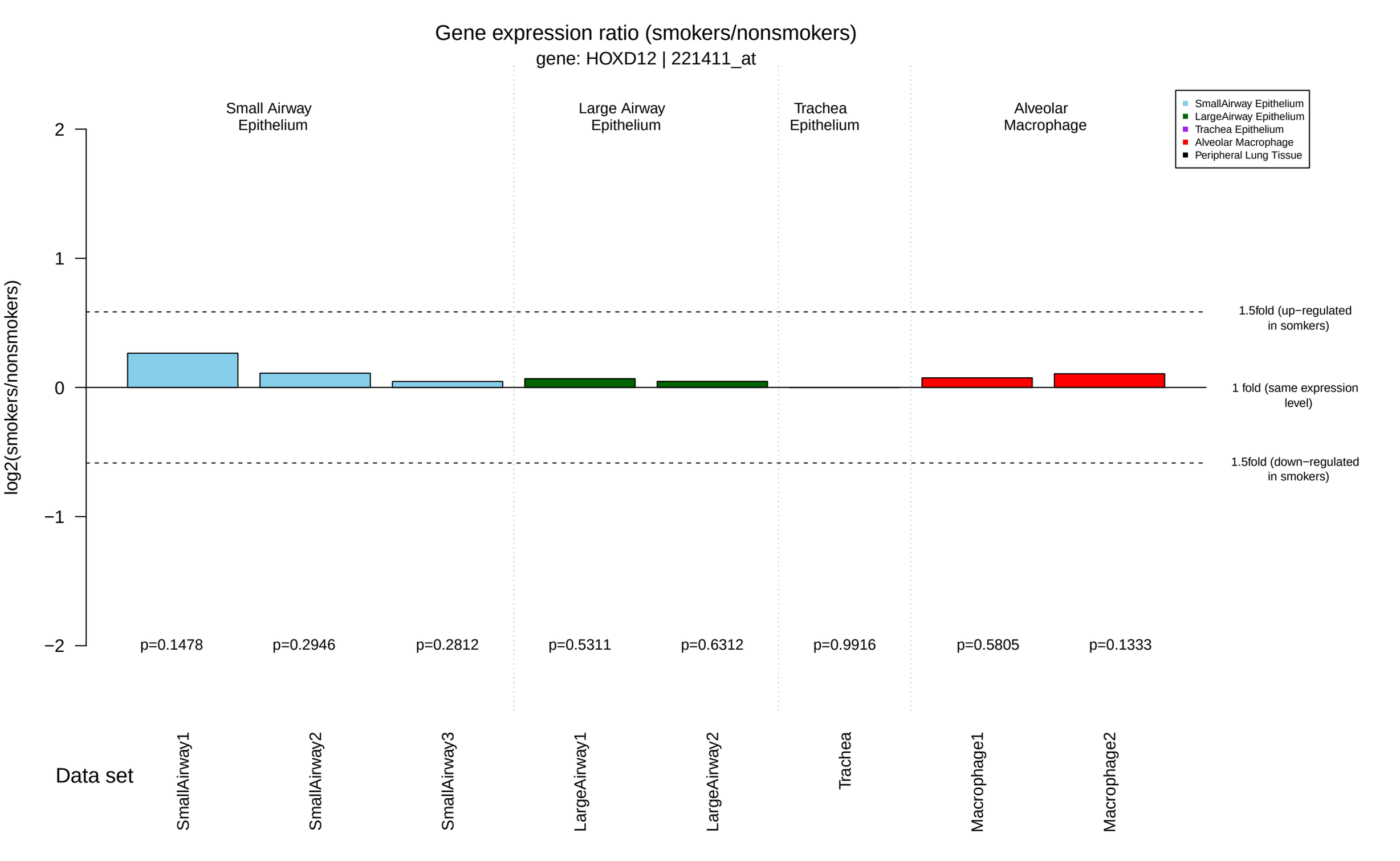
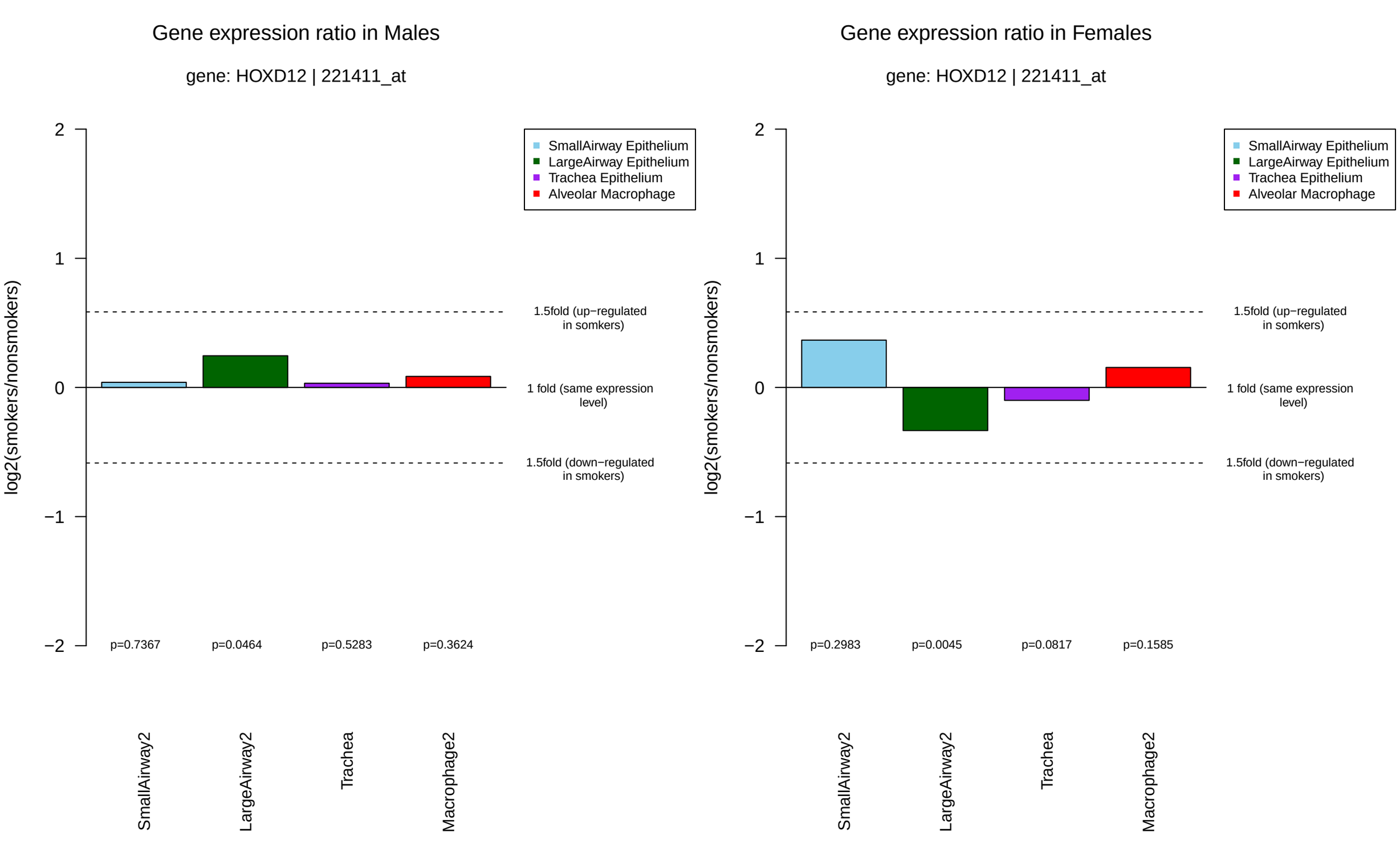
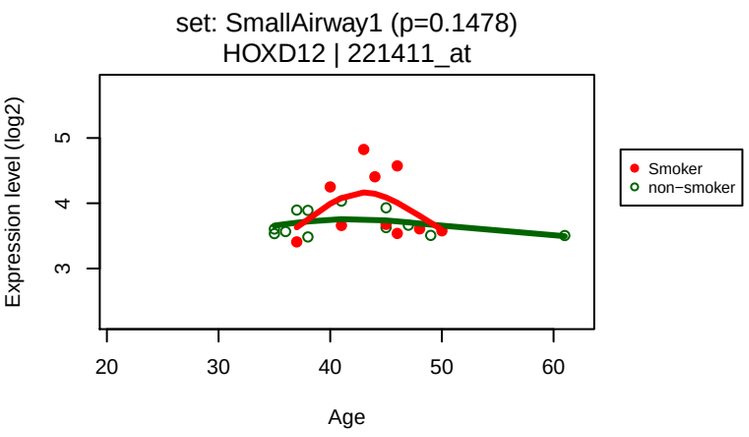
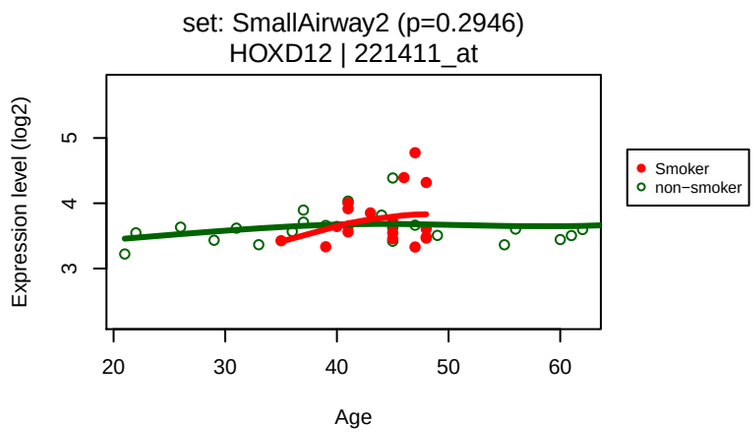
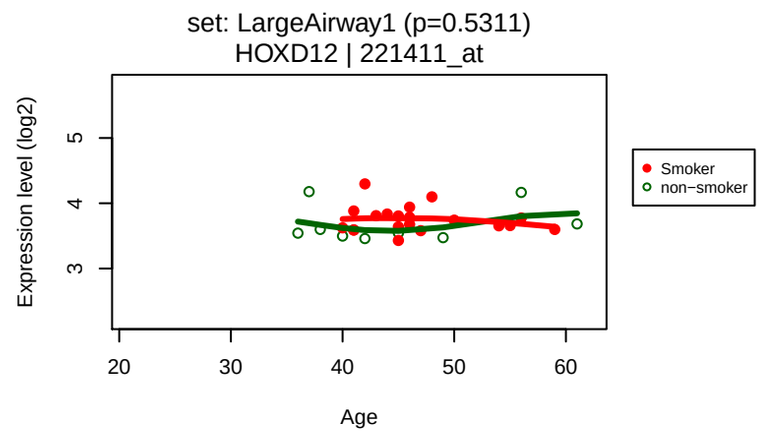

Gene expression of smokers and nonsmokers
gene: HOXD12 | 221411_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1478
p=0.2946
p=0.2812
p=0.5311
p=0.6312
p=0.9916
p=0.5805
p=0.1333
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

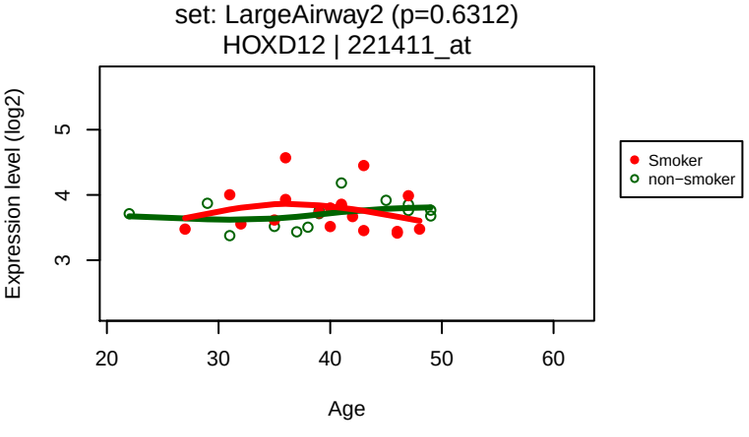

Gene expression ratio (smokers/nonsmokers)
gene: HOXD12 | 221411_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1478
p=0.2946
p=0.2812
p=0.5311
p=0.6312
p=0.9916
p=0.5805
p=0.1333
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

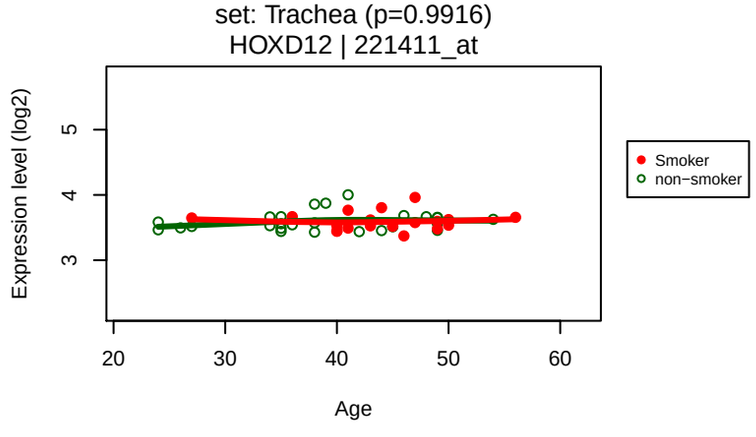

Gene expression ratio in Males
gene: HOXD12 | 221411_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7367
p=0.0464
p=0.5283
p=0.3624
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HOXD12 | 221411_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2983
p=0.0045
p=0.0817
p=0.1585
SmallAirway2
LargeAirway2
Trachea
Macrophage2

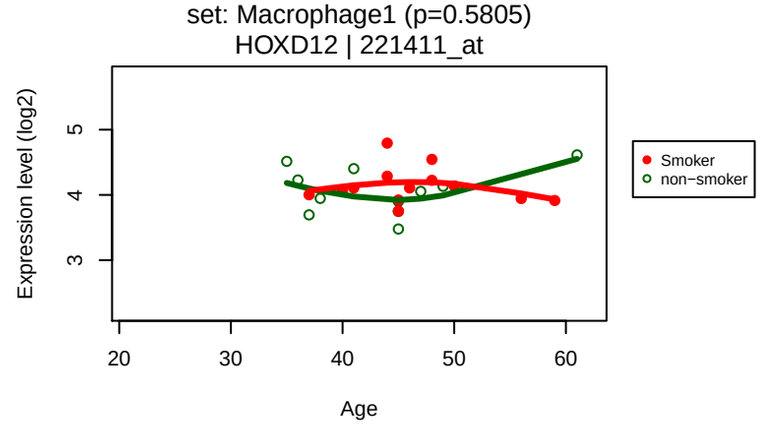

set: SmallAirway1 (p=0.1478)
HOXD12 | 221411_at
Expression level (log2)
Age
Smoker
non−smoker

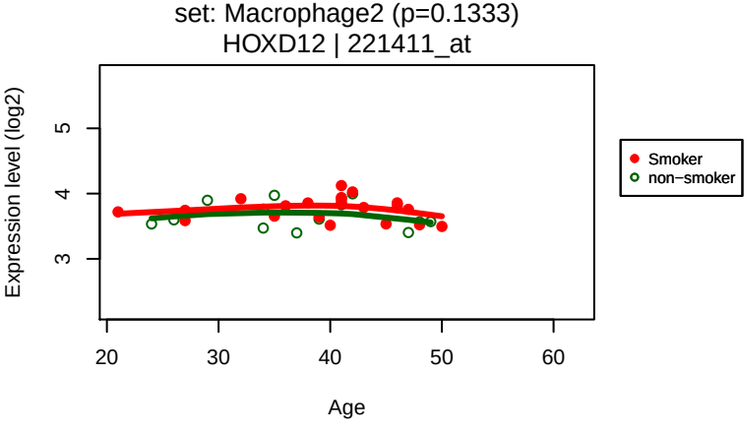

set: SmallAirway2 (p=0.2946)
HOXD12 | 221411_at
Expression level (log2)
Age
Smoker
non−smoker

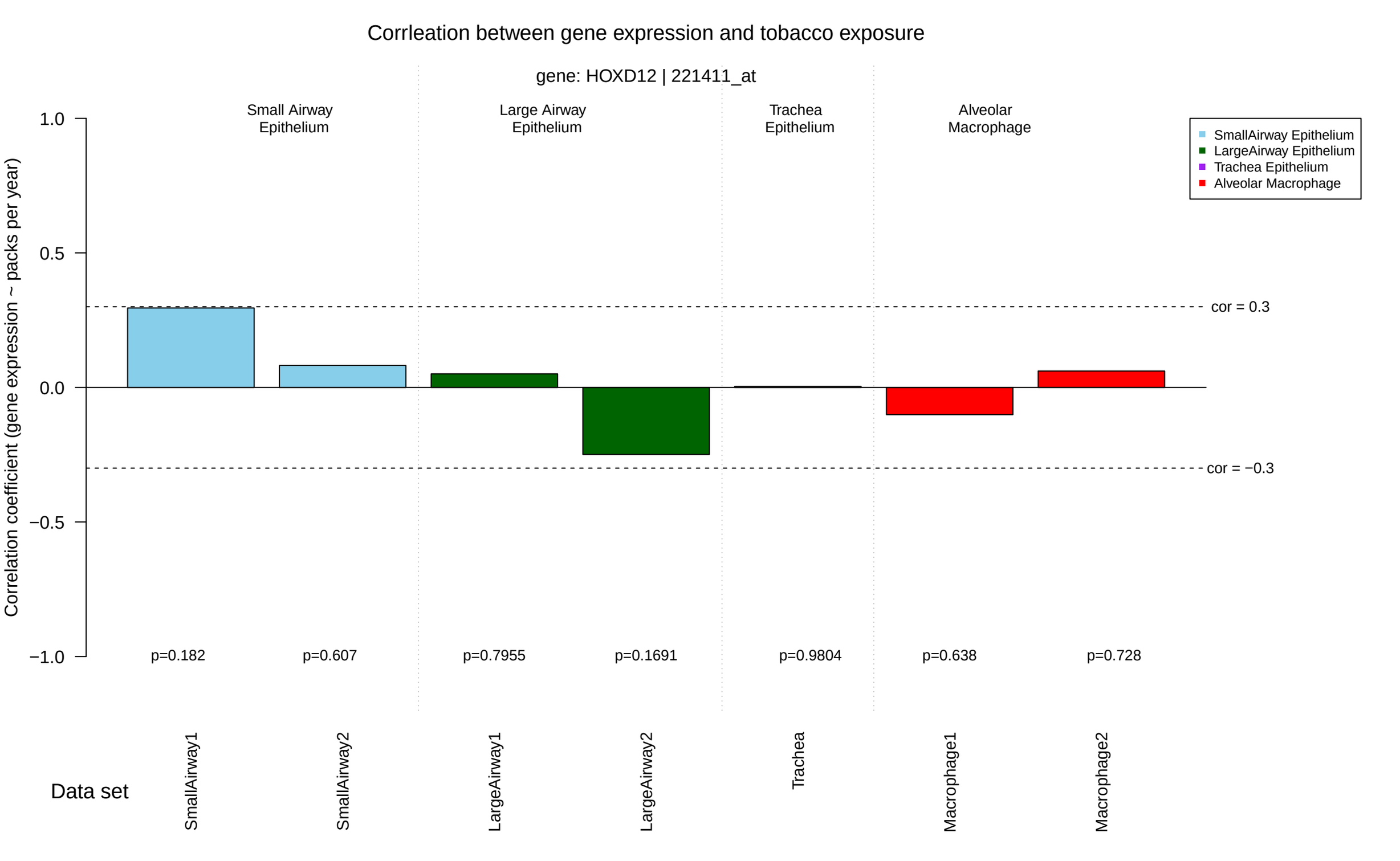

set: LargeAirway1 (p=0.5311)
HOXD12 | 221411_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6312)
HOXD12 | 221411_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9916)
HOXD12 | 221411_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5805)
HOXD12 | 221411_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1333)
HOXD12 | 221411_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HOXD12 | 221411_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.182
p=0.607
p=0.7955
p=0.1691
p=0.9804
p=0.638
p=0.728
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2